

HW3 Solution

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Consider the study of the effect of ploidy on the survival of patients with cancer of the tongue. Using the data on aneuploid tumors.

(a) Estimate the survival function. Use the KM estimator.

讀入資料

```
#install.packages("KMsurv") #安裝套件
library(KMsurv)
data(tongue)
```

從中挑選出 aneuploid tumors 的病人

```
tongue.1<-tongue[tongue$type==1,]
```

利用 *survfit* 指令得到 KM estimator

```
library(survival)

r.km<-survfit(Surv(time,delta)~1,data=tongue.1)
summary(r.km)
```

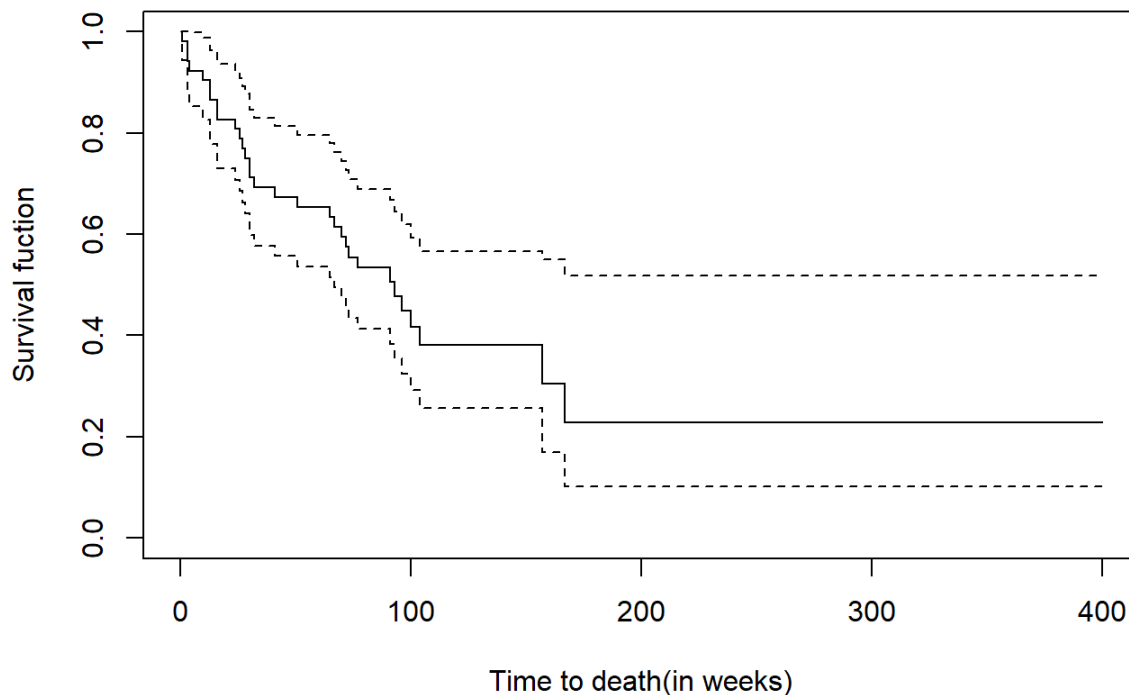
```
## Call: survfit(formula = Surv(time, delta) ~ 1, data = tongue.1)
##
##      time n.risk n.event survival std.err lower 95% CI upper 95% CI
##      1      52      1   0.981  0.0190   0.944   1.000
##      3      51      2   0.942  0.0323   0.881   1.000
##      4      49      1   0.923  0.0370   0.853   0.998
##     10      48      1   0.904  0.0409   0.827   0.988
##     13      47      2   0.865  0.0473   0.777   0.963
##     16      45      2   0.827  0.0525   0.730   0.936
##     24      43      1   0.808  0.0547   0.707   0.922
##     26      42      1   0.788  0.0566   0.685   0.908
##     27      41      1   0.769  0.0584   0.663   0.893
##     28      40      1   0.750  0.0600   0.641   0.877
##     30      39      2   0.712  0.0628   0.598   0.846
##     32      37      1   0.692  0.0640   0.578   0.830
##     41      36      1   0.673  0.0651   0.557   0.813
##     51      35      1   0.654  0.0660   0.537   0.797
##     65      33      1   0.634  0.0669   0.516   0.780
##     67      32      1   0.614  0.0677   0.495   0.762
##     70      31      1   0.594  0.0683   0.475   0.745
##     72      30      1   0.575  0.0689   0.454   0.727
##     73      29      1   0.555  0.0693   0.434   0.709
##     77      27      1   0.534  0.0697   0.414   0.690
##     91      19      1   0.506  0.0715   0.384   0.667
##     93      18      1   0.478  0.0728   0.355   0.644
##     96      16      1   0.448  0.0741   0.324   0.620
##    100      14      1   0.416  0.0754   0.292   0.594
##    104      12      1   0.381  0.0767   0.257   0.566
##    157       5      1   0.305  0.0918   0.169   0.550
##    167       4      1   0.229  0.0954   0.101   0.518
```

根據 time 及 survival 欄位，可得到 survival function 的 KM estimator ($S(t)$) 為

index	time interval	S(t)
1	$0 \leq t < 1$	1.000
2	$1 \leq t < 3$	0.981
3	$3 \leq t < 4$	0.942
4	$4 \leq t < 10$	0.923
5	$10 \leq t < 13$	0.904
6	$13 \leq t < 16$	0.865
7	$16 \leq t < 24$	0.827
8	$24 \leq t < 26$	0.808
9	$26 \leq t < 27$	0.788
10	$27 \leq t < 28$	0.769
11	$28 \leq t < 30$	0.750
12	$30 \leq t < 32$	0.712
13	$32 \leq t < 41$	0.692
14	$41 \leq t < 51$	0.673
15	$51 \leq t < 65$	0.654
16	$65 \leq t < 67$	0.634
17	$67 \leq t < 70$	0.614
18	$70 \leq t < 72$	0.594
19	$72 \leq t < 73$	0.575
20	$73 \leq t < 77$	0.555
21	$77 \leq t < 91$	0.534
22	$91 \leq t < 93$	0.506
23	$93 \leq t < 96$	0.478
24	$96 \leq t < 100$	0.448
25	$100 \leq t < 104$	0.416
26	$104 \leq t < 157$	0.381
27	$157 \leq t < 167$	0.305
28	$167 \leq t < 400$	0.229

(b) plot the estimated survival curve.

```
plot(r.km, xlab = "Time to death(in weeks)", ylab = "Survival fuction")
```



(c) Estimate the survival function at 12 weeks and 60 weeks after transplant. Find the standard errors for your estimates.

```
#S(12)
#round(summary(r.km)$surv[sum(summary(r.km)$time<=12)],digits=3)
#round(summary(r.km)$std.err[sum(summary(r.km)$time<=12)],digits=3)
summary(r.km,12)
```

```
## Call: survfit(formula = Surv(time, delta) ~ 1, data = tongue.1)
##
##   time n.risk n.event survival std.err lower 95% CI upper 95% CI
##    12    47      5    0.904  0.0409    0.827    0.988
```

```
#S(60)
#round(summary(r.km)$surv[sum(summary(r.km)$time<=60)],digits=3)
#round(summary(r.km)$std.err[sum(summary(r.km)$time<=60)],digits=3)
summary(r.km,60)
```

```
## Call: survfit(formula = Surv(time, delta) ~ 1, data = tongue.1)
##
##   time n.risk n.event survival std.err lower 95% CI upper 95% CI
##    60    34     18    0.654  0.066    0.537    0.797
```

$\hat{S}(12) = 0.904$, standard error $se(\hat{S}(12)) = 0.041$

$\hat{S}(60) = 0.654$, standard error $se(\hat{S}(60)) = 0.066$

(d) Estimate the cumulative hazard function. Use the Nelson-Aalen estimator.

```
(r.cumhaz<-data.frame(r.km$time,r.km$n.risk,r.km$n.event,r.km$cumhaz)[r.km$n.event>0,])
```

##	r.km.time	r.km.n.risk	r.km.n.event	r.km.cumhaz
## 1	1	52	1	0.01923077
## 2	3	51	2	0.05844646
## 3	4	49	1	0.07885462
## 4	10	48	1	0.09968795
## 5	13	47	2	0.14224114
## 6	16	45	2	0.18668559
## 7	24	43	1	0.20994140
## 8	26	42	1	0.23375093
## 9	27	41	1	0.25814117
## 10	28	40	1	0.28314117
## 11	30	39	2	0.33442322
## 12	32	37	1	0.36145025
## 13	41	36	1	0.38922803
## 14	51	35	1	0.41779945
## 16	65	33	1	0.44810248
## 17	67	32	1	0.47935248
## 18	70	31	1	0.51161055
## 19	72	30	1	0.54494388
## 20	73	29	1	0.57942664
## 22	77	27	1	0.61646368
## 29	91	19	1	0.66909526
## 30	93	18	1	0.72465081
## 31	96	16	1	0.78715081
## 33	100	14	1	0.85857938
## 35	104	12	1	0.94191272
## 41	157	5	1	1.14191272
## 42	167	4	1	1.39191272

根據 time 及 cumhaz 欄位，可得到 cumulative hazard function 的 Nelson-Aalen estimator ($\tilde{\Lambda}(t)$) 為

index	time interval	Lambda(t)
1	0<=t<1	0.000
2	1<=t<3	0.019
3	3<=t<4	0.058
4	4<=t<10	0.079
5	10<=t<13	0.100
6	13<=t<16	0.142
7	16<=t<24	0.187
8	24<=t<26	0.210
9	26<=t<27	0.234
10	27<=t<28	0.258
11	28<=t<30	0.283
12	30<=t<32	0.334
13	32<=t<41	0.361
14	41<=t<51	0.389
15	51<=t<65	0.418
16	65<=t<67	0.448
17	67<=t<70	0.479

index	time interval	Lambda(t)
18	70<=t<72	0.512
19	72<=t<73	0.545
20	73<=t<77	0.579
21	77<=t<91	0.616
22	91<=t<93	0.669
23	93<=t<96	0.725
24	96<=t<100	0.787
25	100<=t<104	0.859
26	104<=t<157	0.942
27	157<=t<167	1.142
28	167<=t<400	1.392

(e) Estimate the cumulative hazard rate at 60 weeks and find the standard error. Estimate the survival rate at 60 weeks by using the cumulative hazard rate and compare to your estimate in (c).

```
(cumhaz.60<-summary(r.km,60)$cumhaz)
```

```
## [1] 0.4177995
```

```
summary(r.km,60)$std.chaz
```

```
## [1] 0.09921313
```

cumulative hazard rate 估計值為 $\tilde{\Lambda}(60) = 0.4178$, 估計值的 standard error 為 0.0992

```
(s.60.na<-exp(-cumhaz.60))
```

```
## [1] 0.6584943
```

$$\tilde{S}(60) = \exp\{-\tilde{\Lambda}(60)\} = 0.6585$$

使用 KM estimator 存活率估計值為 0.654，和利用 Nelson-Aalen 所得的存活率估計值 相近。